

Causal Inference Lab 1: Introduction & Probability Review

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¹Many examples influenced by Probability and Statistics for Data Science by
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A little About Me!

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Office Hours: Tuesdays, 12PM - 2PM at CDS (Room 765)
- ▶ **Education:**
 - ▶ M.S. in Data Science at CDS
 - ▶ B.A. in Economics (Minors in Data Science and Mathematics)
- ▶ The **goal of the lab** is to reinforce the material through application, real-world cases, and examples.

A Little About You?

- ▶ Name
- ▶ Major, interests, where you're from, etc?

Content

- ① Introduction
- ② Probability Review
 - ① Expectations
 - ② Variance

Introduction

The Philosophy of Causal Inference

The World of Prediction (Before)

- ▶ **Goal:** *predict* an outcome Y from variables X .
- ▶ **Example:** Use age, income, and clicks to predict ad response.

$$Y = \beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{income} + \beta_3 \cdot \text{clicks} + \varepsilon$$

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The World of Causation (Now)

- ▶ **Goal:** *understand and identify* how X affects Y .
- ▶ Not just prediction: "*What if we intervene on X ?*"
- ▶ Requires probability, statistics, and counterfactuals.

Types of Causal Questions (Among others)

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- ➎ **Heterogeneity of Causation:** Does X affect some groups more than others?
 - ▶ e.g., Do job training programs help low-income workers more than high-income workers?

The Fundamental Problem of Causal Inference

Example: Evaluating a Drug (From Probability and Statistics for Data Science)

- ▶ **Goal:** Evaluating the efficacy of a drug (i.e., does it help patients recover?).
- ▶ **Intervention:** Patients either receive the drug ($T = 1$) or not ($T = 0$).

$$T = \begin{cases} 1, & \text{if patient receives the drug} \\ 0, & \text{if patient does not receive the drug} \end{cases}$$

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- ▶ **Outcome:** Patients either recover ($Y = 1$) or not ($Y = 0$).

$$Y = \begin{cases} 1, & \text{if patient recovers} \\ 0, & \text{if patient does not recover} \end{cases}$$

- ▶ **Question:** How can we estimate the effectiveness of the drug?

The Fundamental Problem of Causal Inference

Naive Approach

- **Natural Idea:** compare recovery rates between treated and untreated groups:

The Fundamental Problem of Causal Inference

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- **Problem:** This is only a **correlation**, not necessarily a **causal effect**.
 - Treated and untreated patients may differ in many other ways.
 - e.g., Sicker patients might be more likely to get the drug.
- This motivates the **Fundamental Problem of Causal Inference**:

The Fundamental Problem of Causal Inference cont.

- ▶ To know if a treatment *causes* an effect, we would like to compare:
 - ▶ The outcome if the group **receives treatment**.
 - ▶ The outcome if the **same group does not receive treatment**.
- ▶ But in reality:
 - ▶ We only ever see one of these outcomes.
 - ▶ We cannot observe both worlds at the same time.

Fundamental Problem of Causality

We cannot observe the same patient both with treatment ($T = 1$) and without treatment ($T = 0$).

- ▶ **Therefore**, we must be creative in methodology: randomized experiments, natural experiments, statistical adjustments.

The Need for Probability (and Statistics) in Causal Inference

Causal inference is built on two pillars:

1 Probability: Understanding Uncertainty

- ▶ Provides a formal language to describe randomness and uncertainty.
- ▶ Lets us reason about what we would expect to see under different causal models.
- ▶ Example: If smoking really caused cancer, what would be the probability of observing cancer rates in smokers vs. non-smokers?

2 Statistics: Learning from Data

- ▶ Provides tools to estimate causal quantities from real-world data.
- ▶ Allows us to test whether observed differences are likely due to chance or reflect true causal effects.
- ▶ Example: Using sample data to estimate how much education increases earnings.

Probability Review²

- ▶ **Probability** provides the foundation for reasoning under uncertainty.
- ▶ In this section, we will briefly review:
 - ▶ Random variables (discrete and continuous)
 - ▶ Key functions: PMF, PDF, and CDF
 - ▶ Expected Value
 - ▶ Variance (and Standard Deviation)
- ▶ **Goal:** Make sure everyone is comfortable with the basics before moving into causal inference.

²This is *actually* a review – there is an expectation that you have encountered this material before

What is a Random Variable (RV)?

- ▶ **Random Variable (RV):** A *function* that *maps* all elements of a sample space to the real numbers
 - ▶ **Mathematically:** $\Omega \rightarrow \mathbb{R}$
 - ▶ **RV Notation:** $\mathcal{X}, X, \tilde{x}$
- ▶ **Intuition:** Sometimes, the sample space is not represented in numbers
 - ▶ The role of RVs is to map all those elements in the sample space Ω (that are, at times, not numbers) into real numbers $\mathbb{R}$
- ▶ **Relevance:**
 - 1 We use RVs to describe probability distributions
 - 2 We use RVs to deal with probabilities

Random Variables as Functions?

Example: Flipping a Coin ($\Omega \rightarrow \mathbb{R}$)

- ▶ **Scenario:** You flip a coin and want to reason about the outcome quantitatively.
- ▶ **Sample Space:**

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- ▶ Define a Random Variable X that "transforms" the elements in the sample space to real numbers:

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- ▶ Define a Random Variable X that "transforms" the elements in the sample space to real numbers:

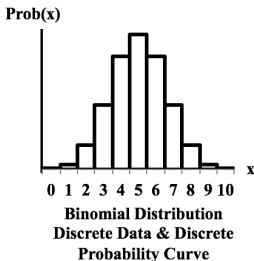
$$X(\omega) := \begin{cases} 0, & \text{if } \omega = \text{Heads} \\ 1, & \text{if } \omega = \text{Tails} \end{cases}$$

- ▶ **In words:** The RV X takes on the value 0 if the outcome is Heads and 1 if Tails
 - ▶ There you go! We have successfully mapped outcomes from the sample space Ω to real numbers $\mathbb{R}$

Two Types of Random Variables (RVs)

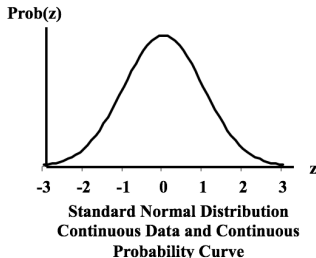
(1) Discrete

- ▶ Countable finite numbers of values
- ▶ **Example:** Goals scored in a soccer match



(2) Continuous

- ▶ Can take an infinite number of values, intervals
- ▶ **Example:** Marathon time



Two Types of Random Variables

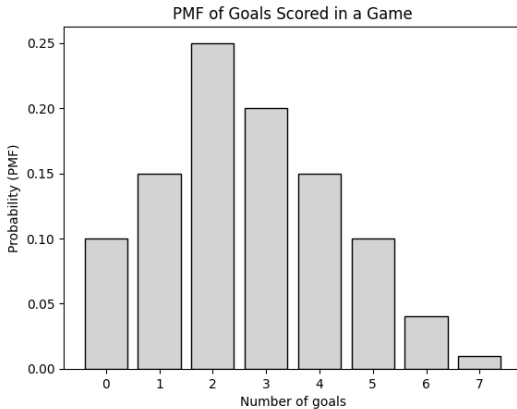
(1) Discrete vs. (2) Continuous

Discrete RV	Continuous RV
Takes values from a <i>countable</i> set	Takes values from an <i>uncountable</i> set (interval)
Probability Mass Function (PMF): $\mathbb{P}(X = x) = p(x), \quad \sum_x p(x) = 1$	Probability Density Function (PDF): $f(x), \quad \int_{-\infty}^{\infty} f(x) dx = 1$
Cumulative Distribution Function (CDF): $F(x) = \mathbb{P}(X \leq x) = \sum_{t \leq x} p(t)$	Cumulative Distribution Function (CDF): $F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) dt$
Example: Number of heads in 3 coin flips	Example: Distribution of student grades

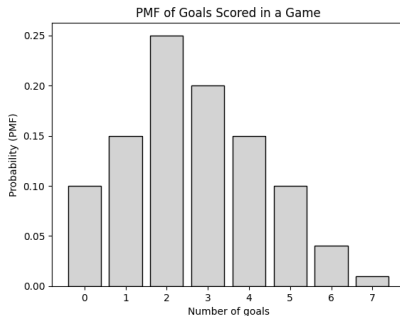
Discrete Random Variables

Example: Goals Scored

- ▶ Let G be the random variable denoting the number of goals scored.
- ▶ G can take values in $\{0, 1, 2, \dots, 7\}$ (based on our dataset).

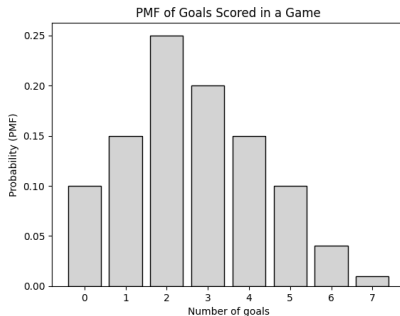


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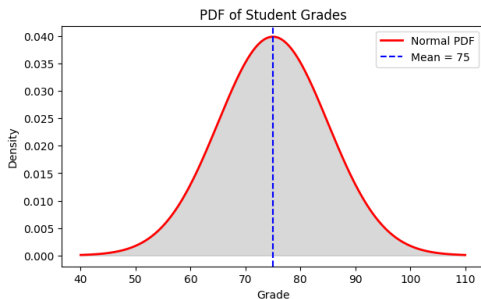
$$P(G = 1 \text{ or } G = 7) = P(G = 1) + P(G = 7) = 0.15 + 0.01 = 0.16$$

Continuous Random Variables

Example: Student Grades

- ▶ Let G be the random variable denoting the grade of a student.
- ▶ G is modeled as a *continuous* variable with values in $[0, 100]$.
- ▶ The distribution of G is approximately Normal with mean $\mu = 75$ and standard deviation $\sigma = 10$:

$$G \sim \mathcal{N}(75, 10^2)$$



Expected Value

Expected value

- ▶ The **Expected Value** (or Expectation) is the theoretical average ("long-run typical value") of a Random Variable.
- ▶ It is a **weighted average** of all possible values of the random variable.

- ▶ **Expected Value of a Discrete RV:**

$$\mathbb{E}[X] = \sum_x x P(X = x)$$

- ▶ **Expected Value of a Continuous RV:**

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Expected Value

Difference between Expected Value and the Mean?

- **Expected Value ($\mathbb{E}[X]$):** Theoretical average of a random variable, defined by the probability model.

$$\mathbb{E}[X] = \sum_x xP(X=x) \quad \text{or} \quad \int_{-\infty}^{\infty} xf(x)dx$$

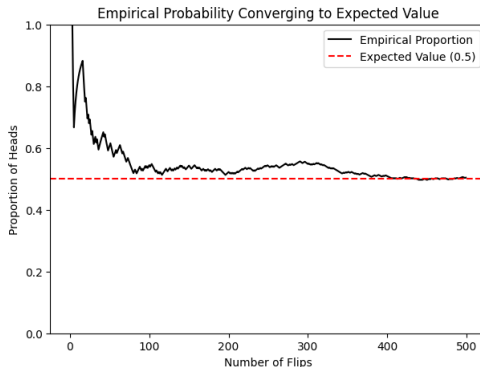
- **Sample Mean ($\bar{x}$):** Average computed from observed data.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Key Point:** With finite data, $\bar{x}$ may differ from $\mathbb{E}[X]$. As $n \rightarrow \infty$, $\bar{x} \rightarrow \mathbb{E}[X]$.

Expected Value

Example: Expected Value vs Sample Mean



- Observe how the empirical proportion (the *sample mean*) converges to the expected value (the *theoretical average*) as the sample size (number of flips) increases. This illustrates the **Law of Large Numbers**.

Expected Value

Example with One Random Variable

g	0	1	2	3	4	5	6	7
$P(G = g)$	0.10	0.15	0.25	0.20	0.15	0.10	0.04	0.01

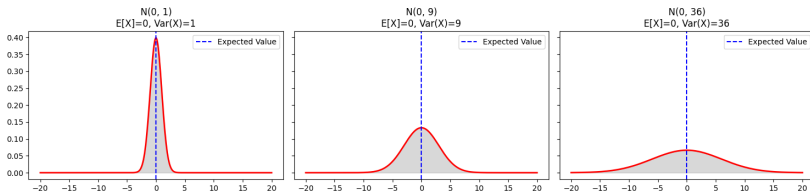
$$\mathbb{E}[G] = \sum_{g=0}^7 g \cdot P(G = g)$$

$$\begin{aligned} &= 0(0.10) + 1(0.15) + 2(0.25) + 3(0.20) + 4(0.15) + 5(0.10) + 6(0.04) + 7(0.01) \\ &= 0 + 0.15 + 0.50 + 0.60 + 0.60 + 0.50 + 0.24 + 0.07 \\ &= 2.66 \end{aligned}$$

Variance

- **Expected value** is one measure to characterize a random variable

Same Expected Value, Different Spread



- Observe how random variables can have the same expected value, but greatly vary in distribution.
- This inspires the need for a **second measure**, which we call **Variance**

Variance cont.

Variance

Variance measures how much a random variable deviates from its expected value.

$$\begin{aligned} \underbrace{E[(X - E[X])^2]}_{\text{Variance}} &= E[X^2 - 2XE[X] + E[X]^2] \\ &= E[X^2] - E[2XE[X]] + E[X]^2 \\ &= E[X^2] - 2E[X]E[X] + E[X]^2 \\ &= \boxed{E[X^2] - E[X]^2} \end{aligned}$$

- **Interpretation:** Variance quantifies the *spread* or *dispersion* of the distribution around the mean.
- The square root of variance is the **standard deviation**, which is in the same units as the data.